Bacterial Growth and Culture

Microbiology - Lecture 4

Department of Medical Microbiology

Objectives

- Describe the bacterial growth curve
- Classify bacteria by their temperature and oxygen needs
- Describe common culture media

Bacterial Growth

- Bacteria reproduce by binary fission: one cell divides into two identical daughter cells.
- Generation time is the time a population needs to double; for Escherichia coli under ideal conditions it is about 20 minutes.

The Growth Curve

- Lag phase: the cells adapt to the new medium and make enzymes, but their number hardly changes.
- Log (exponential) phase: the cells divide at a constant, maximal rate and are most sensitive to antibiotics.
- Stationary phase: nutrients run out and waste builds up, so the number of new cells equals the number of dying cells.
- Death phase: the number of living cells declines.

Temperature Requirements

- Psychrophiles grow best below about 15 degrees Celsius.
- Mesophiles grow best between about 20 and 45 degrees Celsius; most human pathogens are mesophiles with an optimum near 37 degrees.
- Thermophiles grow best above about 45 degrees Celsius.

Oxygen Requirements

- Obligate aerobes require oxygen for growth.
- Obligate anaerobes are killed by oxygen; Clostridium species are examples.
- Facultative anaerobes grow with or without oxygen but grow better with it; Escherichia coli is an example.
- Microaerophiles need oxygen at a lower concentration than in air.

Culture Media

- Selective media contain substances that inhibit some organisms and let others grow.
- Differential media let different organisms be told apart by their appearance, such as the colour of their colonies.
- MacConkey agar is both selective and differential: it inhibits Gram-positive bacteria and separates lactose fermenters (pink colonies) from non-fermenters.
- On blood agar, beta-haemolysis is a clear zone around the colony, alpha-haemolysis a greenish zone, and gamma means no haemolysis.

Measuring Growth

- A viable plate count counts only living cells and reports colony-forming units (CFU).
- Turbidity measured with a spectrophotometer estimates living and dead cells together.

Check your understanding

1. In which phase would an antibiotic that acts on the cell wall work best?
2. Pink colonies grow on MacConkey agar. What do they tell you?

Thank you